

JINGZHANG SIGNAL LINE 京张信号带

Translating a century of railway signalling discipline into a governance and spatial protocol for the AI innovation belt · A3 booklet (conceptual)

In 1909 the Jing-Zhang Railway opened — the first trunk railway built independently by Chinese engineers; a century of railway operations answered “how to run safely” with signalling discipline. This proposal translates red-yellow-green grading, route interlocking, manual go-live confirmation, and fail-safe operation into a governance and spatial protocol for the AI innovation belt.

All spatial proposals are conceptual, based on the provisional rough boundary (not an official redline), for professional teams to deepen; they do not replace statutory planning or constitute government review conclusions. Recalculate package-wide when official polygons are released.

1 Design basis and source list

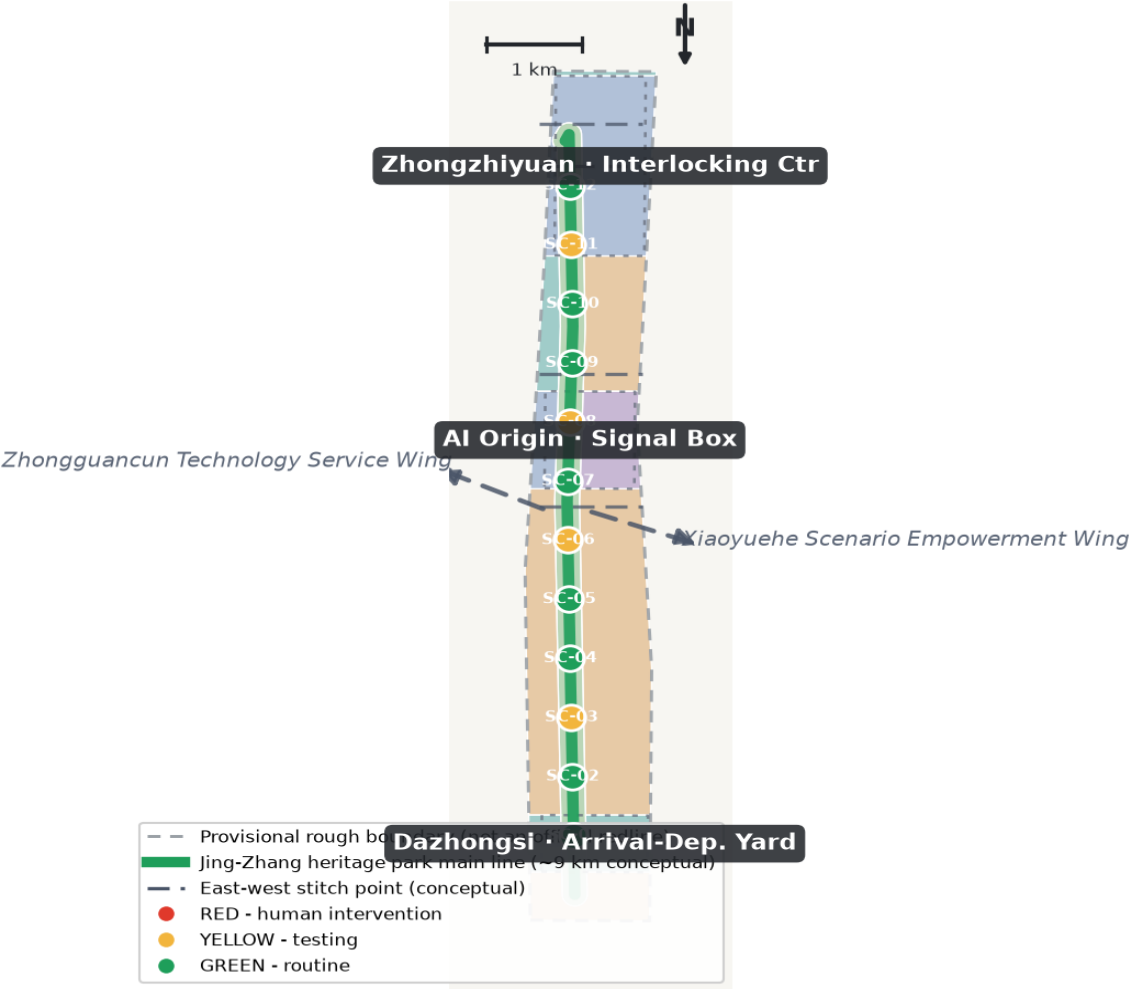
Primary basis: the Qualification Pre-announcement (Haidian Branch, 2026-05-09); the agent open-call taskbook (cleared) adds six tasks and ten co-creation principles; professional standards follow the Measures for the Administration of Urban Design and the Regulatory-Detailed-Planning Measures. Full provenance in sources.json.

2 Three-level scope framework

Level	Area	Task
Coordinated research	43.6 km²	Ecosystem & future city form
Overall design	11.4 km²	Renewal, land, transport, character
Key areas	368.4 ha	Detailed design of 3 areas

3 Master concept: one line, three stations, two wings, twelve nodes

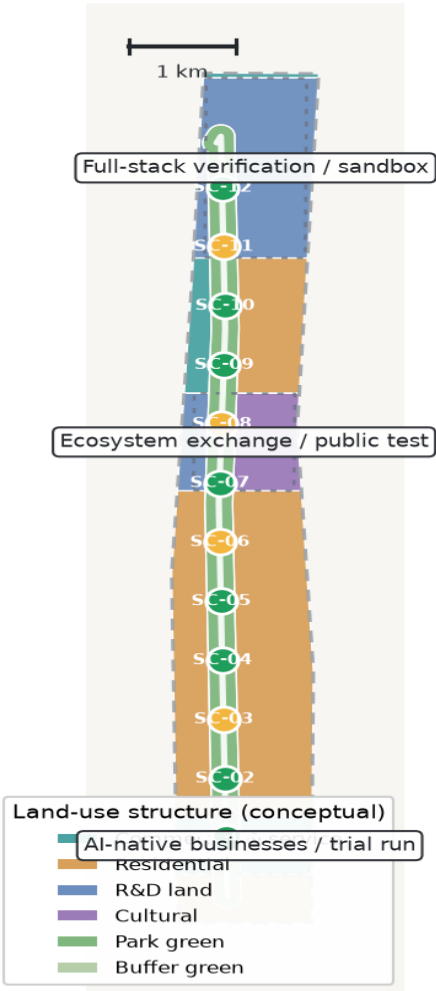
JINGZHANG SIGNAL LINE · One line, three stations, two wings, twelve nodes



Source: generated geometry GeoJSON (recomputed EPSG:4548, output 4326) · conceptual, pending official data · generated 2026-08-11 v1.1

4 Land-use & spatial structure (conceptual)

Land-use and spatial structure (conceptual, EPSG:4548)



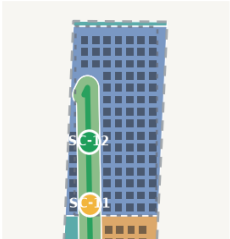
Source: generated geometry GeoJSON (recomputed EPSG:4548, output 4326) · conceptual, pending official data · generated 2026-08-11 v1.1

Park green 232.1 ha (20.3%) · R&D; 208.8 ha (18.3%) · residential 531.0 ha (46.5%) · cultural 66.9 ha · commercial+business 58.5 ha · station 44.0 ha — intent layer, not an existing-condition survey or statutory land-use plan.

5 Detailed design of the three key areas

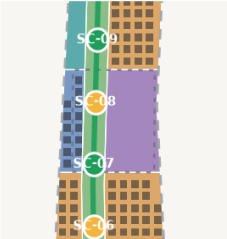
Detailed design of the three key areas (conceptual)

Zhongzhiyuan · Interlocking Center



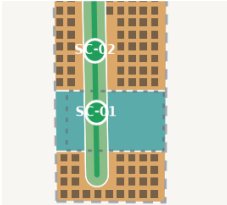
Central verification ground & interlocking sandboxes / R&D clusters / Interlocking Hall (manual cognitive zone)

AI Origin · Signal Box



Origin Plaza & Zero Signal Tower (pilgrimage landmark) / public test ground / campus slow-mobility stitching

Dazhongsi · Arrival-Departure Yard



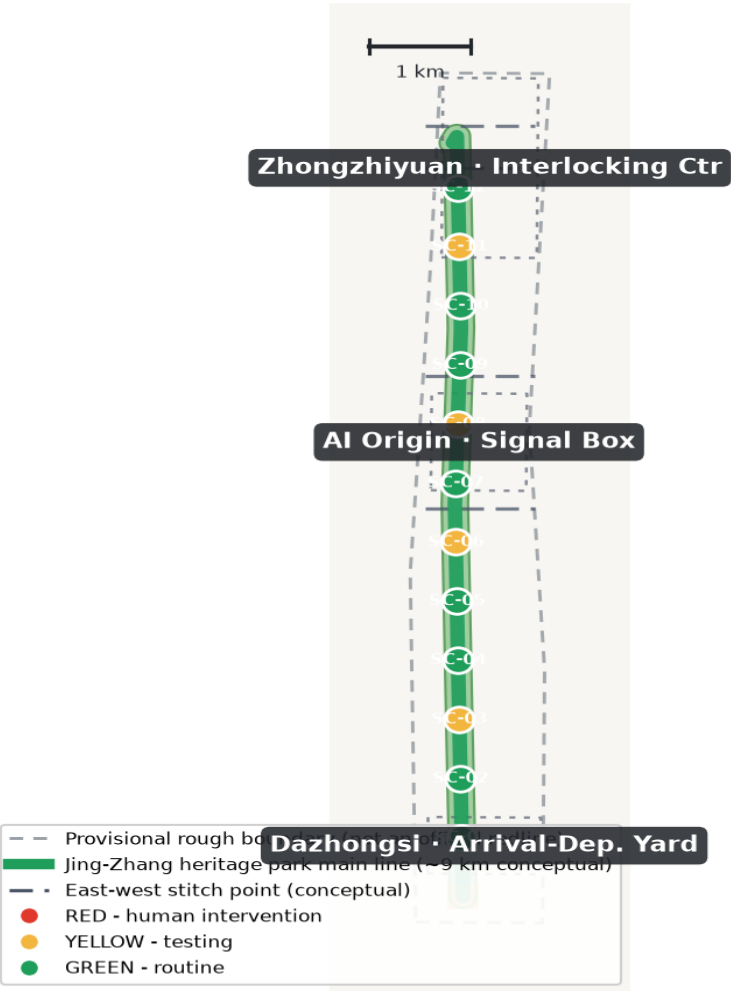
Trial-run block (yellow→review→green/offline) / Arrival Bell / TOD linkage

Source: generated geometry GeoJSON (recomputed EPSG:4548, output 4326) · conceptual, pending official data

Zhongzhiyuan · Interlocking Center (192.9 ha): verification ground & sandboxes, R&D; clusters, Interlocking Hall; AI Origin · Signal Box (104.3 ha): Origin Plaza & Zero Signal Tower, public test ground, campus stitching; Dazhongsi · Arrival-Departure Yard (72.0 ha): trial-run block, Arrival Bell, TOD linkage.

6 Slow mobility and blue-green network

Slow mobility · rail connection · blue-green network (conceptual)



Source: generated geometry GeoJSON (recomputed EPSG:4548, output 4326) · conceptual, pending official data · generated 2026-08-11 v1.1

7 AI scenarios and the signal protocol

ID	Scenario	Signal
SC-01	Slow-mobility navigation	G
SC-02	Cultural AI guide	G
SC-03	Origin AI service desk	G
SC-04	Smart consumption block	G
SC-05	Park robot patrol (test)	Y
SC-06	Edge compute station	G
SC-07	AI time-sync desk	G
SC-08	AI pilgrimage line	G
SC-09	Manual fallback points	G
SC-10	Interlocking sandbox (test 1)	Y
SC-11	Public test ground (test 2)	Y
SC-12	Trial-run line (test 3)	Y

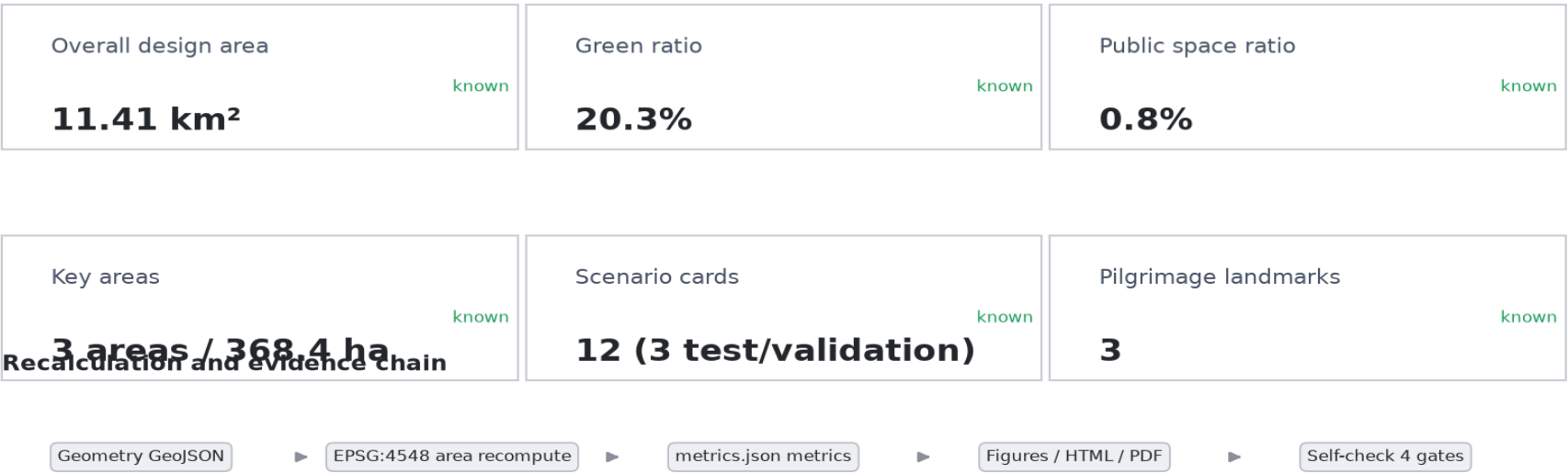
Red-light protocol (institutional base): any AI service degrades to human on failure with public record; green = routine with periodic review; yellow = offline if not cleared by deadline.

8 Renewal projects, phasing, compliance

Projects UP-01 main-line stitching · UP-02 Origin Plaza & Zero Tower · UP-03 sandbox · UP-04 public test ground · UP-05 trial-run block · UP-06 compute stations · UP-07 Arrival Bell · UP-08 stitch points. Phasing: 1 spine+Origin (308 ha) → 2 Interlocking Center (193 ha) → 3 Arrival-Dep. Yard (72 ha). Announcement 1.3/1.4/1.5 and agent.1–6 fully covered (compliance_matrix.json).

9 Metrics and recalculation

Metrics evidence · recalculation and self-check (EPSG:4548)



Pending official data: FAR / height / density / green ratio / setback / road redline / ownership (planning_limits.json)

Source: generated geometry GeoJSON (recomputed EPSG:4548, output 4326) · conceptual, pending official data
Self-check: deterministic validation · spatial review · visual packaging · professional evidence (self_check.json)

All area metrics are recomputed from the submitted GeoJSON in EPSG:4548; statutory controls (FAR, height, density) are pending official data and not inferred.
Generated 2026-08-11 · v1.1 · AI-agent original · COMMUNITY-DISPLAY-ONLY

10 Risk, copyright, compliance

Key risks: provisional boundary is not an official redline (recompute on release); missing regulatory/ownership data (pending); cases are mechanism references only; legal citations limited to statutory scope; all statements are open co-creation concepts. Graphics and text are AI-agent originals with open-source fonts; no OSM data, no ODbL obligation.